11 Dependence.

grad Dependence

ref Def VI.1.1. $A \subseteq B$ subring. $x \in B$ is integral over A if x is a root of a monic polynomial with coefficients in A.

Def VI 1.2. $A \subseteq B$ subring. $A \subseteq C \subseteq B$ $C := \{b \in B \mid b \text{ is integral over } A\}$

Then C is called the closed closure of A in B.

1) If $C = A$, then call A integrally closed.

2) If $C = B$, then call that B is integral over A.

3) ~~If~~ A integrally closed $\Leftrightarrow$ A is an integral domain & A is integrally closed in $\mathrm{Frac}(A)$

Prop. VII 1.3. $A \subseteq B$. TFAE:

(1) $x \in B$ is integral over A;

(2) $A[x]$ is a finitely generated A-module

(3) $\exists$ subring $C \subseteq B$. s.t. $A[x] \subseteq C \subseteq B$. and C is a finitely generated A-module

(4). There exists a faithful $A[x]$-module M $(\mathrm{Ann}_{A[x]} M = 0)$ which is finitely generated as an A-module.

pf: (1). $\Rightarrow$ (2). $x \in B$ integral over A. $\exists a_1, \dots, a_n \in A$. s.t. $x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n = 0$

$\Rightarrow A[x]$ is generated by $1, x, \dots, x^{n-1}$

(2) $\Rightarrow$ (3). Take $C = A[x]$.

(3) $\Rightarrow$ (4). Take $M = C$. then $yC = 0 \Rightarrow y \cdot 1 = 0$ thus M is a faithful. $A[x]$-module.

(4) $\Rightarrow$ (1). Consider the $A[x]$-module homomorphism ϕ: $M \longrightarrow M$ $m \longmapsto x \cdot m$

M is fg-A-mod, By Caley-Hamilton thm, we have $\exists a_1, \dots, a_n \in A$, s.t. $x^n + a_1 x^{n-1} + \dots + a_n = 0$ in $\mathrm{End}_A(M)$

And from M is a faithful, we have $x^n + a_1 x^{n-1} + \dots + a_n = 0$ in $A[x]$ ▨

Cora. VII. 1.4. x_i $(1 \le i \le n) \in B$ is integral over A. Then $A[x_1, \dots, x_n]$ is a finitely-generated A-algebra

pf: By induction on n (skipped) ▨

Cora VII. 1.5. The integral closure of A in B is a subring of B containing A.

pf: If $x, y \in \overline{A}$. then $A[x, y]$ is a finitely-generated A-module.

$\Rightarrow A[x \pm y] \subseteq A[x, y]$, $A[xy] \subseteq A[x, y] \overset{(3)}{\Rightarrow} x \pm y, xy \in \overline{A}$ ▨

Example: The integral closure of $\mathbb{Z}$ in the algebraic number field L is called the ring of integers O_L in L.

Def VII.1.6. $f: A \to B$ ring homomorphism (B is an A-algebra).

Then f is integral and B is an ~~A-~~ integral A-algebra.

iff B is integral over $f(A)$

Cora VII 1.7. f is integral, and finite-type, then f is finite.

Cora VII 1.8 $A \subseteq B \subseteq C$ subrings. If B integral over A, C integral over B. Then C integral over A

pf: By restrictions of scalars ▨

Cora VII.1.9. $A \subseteq B$ C is the integral closure of A in B. then C is integrally closed in B

pf: $x \in B$ integral over $C \Rightarrow x$ integral over $A \Rightarrow x \in C$ ▨

2. Going-up property.

Prop. VII.2.1. $A \subseteq B$ rings. B integral over A.

i). $\mathfrak{b} \subseteq B$ ideal. $\mathfrak{a} = \mathfrak{b}^c = A \cap \mathfrak{b}$. then $B/\mathfrak{b}$ integral over $A/\mathfrak{a}$

ii). $S^{-1}B$ integral over $S^{-1}A$.

pf: i) $x^n + a_1 x^{n-1} + \dots + a_n = 0 \pmod{\mathfrak{b}}$ ▨

ii). $\frac{x}{s} \in S^{-1}B \Rightarrow x^n + a_1 x^{n-1} + \dots + a_n = 0 \Rightarrow (\frac{x}{s})^n + \dots + a_n = 0$ ▨

Prop. VII.2.2. $A \subseteq B$ integral domains. B integral over A. Then B is a field iff A is a field

pf: $\Leftarrow$: $0 \neq b \in B$. $\Rightarrow b^n + a_1 b^{n-1} + \dots + a_n = 0 \quad (a_n \neq 0)$

$\Rightarrow b^{-1} = -a_n^{-1}(b^{n-1} + a_1 b^{n-2} + \dots + a_{n-1})$

$\Rightarrow$: $0 \neq x \in A \subseteq B \Rightarrow x^{-1} \in B \Rightarrow (x^{-1})^m + b_1 (x^{-1})^{m-1} + \dots + b_m = 0$

$\Rightarrow x^{-1} = -(b_1 + b_2 x + \dots + b_m x^{m-1}) \in A$ ▨

Prop. VII.2.3. $f: A \to B$ integral. $\mathfrak{q} \in \operatorname{Spec} B$ $\mathfrak{p} = \mathfrak{q} \cap A \in \operatorname{Spec} A$. Then $\mathfrak{q}$ maximal $\Leftrightarrow$ $\mathfrak{p}$ maximal

~~In other words, $f^*: \operatorname{Spec} B \to \operatorname{Spec} A$ is the closed mapping.~~

pf: By VII. 2.1. i). $B/\mathfrak{q} \leftarrow A/(\mathfrak{q} \cap A)$ integral. and $A/(\mathfrak{q} \cap A)$, $B/\mathfrak{q}$ are integral domains

$\Rightarrow$: $\mathfrak{q}$ maximal $\Leftrightarrow$ $\mathfrak{p}$ maximal.

Prop 2.4: $\mathfrak{q}, \mathfrak{q}' \subseteq B$ prime, s.t. $\mathfrak{q} \subseteq \mathfrak{q}'$. and $\mathfrak{q}^c = \mathfrak{q}'^c = \mathfrak{p}$ Then $\mathfrak{q} = \mathfrak{q}'$.

pf: $B_{\mathfrak{p}}$ integral over $A_{\mathfrak{p}}$

$$\begin{array}{ccc} A & \to & A_{\mathfrak{p}} \\ \downarrow & & \downarrow \\ B & \to & B_{\mathfrak{p}} \end{array}$$

we have $\mathfrak{p}^e$ in $A_{\mathfrak{p}}$ is maximal.

and $\mathfrak{q}^e$, $\mathfrak{q}'^e$ in $B_{\mathfrak{p}} \Rightarrow \mathfrak{q}^e \subseteq \mathfrak{q}'^e$.

$\Rightarrow$ $\mathfrak{q}^e \cap A_{\mathfrak{p}} = \mathfrak{q}'^e \cap A_{\mathfrak{p}} = \mathfrak{p}^e$. maximal $\Rightarrow$ $\mathfrak{q}^e$, $\mathfrak{q}'^e$ maximal. $\Rightarrow \mathfrak{q} = \mathfrak{q}'$ ▨

Thm VII. 2.5. $f: A \to B$ integral. Then $f^*: \operatorname{Spec} B \to \operatorname{Spec} A$ is surjective.

pf: $\forall \mathfrak{p} \in \operatorname{Spec} A$. $\Rightarrow B_{\mathfrak{p}}$ integral over $A_{\mathfrak{p}}$ Then $\mathfrak{p}A_{\mathfrak{p}}$ maximal.

$$\begin{array}{ccc} B & \xrightarrow{\beta} & B_{\mathfrak{p}} \supseteq m \\ \uparrow & & \uparrow \\ A & \xrightarrow{\alpha} & A_{\mathfrak{p}} \\ \mathfrak{p} & & m \cap A_{\mathfrak{p}} \end{array}$$

choose $m \subseteq B_{\mathfrak{p}}$ maximal. By (2.3) we have $m \cap A_{\mathfrak{p}}$ maximal in $A_{\mathfrak{p}}$

Then By $A_{\mathfrak{p}}$ is a local ring, we have $m \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$

Then let $\mathfrak{q} = \beta^{-1}(m) \in \operatorname{Spec} B \Rightarrow \mathfrak{q} \cap A = \alpha^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p}$ ▨

(Going-up theorem) $A \subseteq B$ rings B integral over A:

$\mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_m \in \operatorname{Spec} B$ and

$\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m \subsetneq \cdots \subsetneq \mathfrak{p}_n$ $(n>m) \in \operatorname{Spec} A$. s.t. $\forall$ $1 \le i \le m$, $\mathfrak{q}_i \cap A = \mathfrak{p}_i$

Then $\exists$ $\mathfrak{q}_{m+1}, \ldots, \mathfrak{q}_n \in \operatorname{Spec} B$. s.t. $\mathfrak{q}_1 \subsetneq \mathfrak{q}_2 \subsetneq \cdots \subsetneq \mathfrak{q}_n$ & $\mathfrak{q}_i \cap A = \mathfrak{p}_i$, $\forall i$

Pf: By induction: $m=1$. $n=2$. we have $\overline{A} = A/\mathfrak{p}_1$. $\overline{B} = B/\mathfrak{q}_1$,

$\overline{A} \subseteq \overline{B}$. and $\overline{B}$ integral over $\overline{A}$

Then by (2.5) we have $\exists$ $\overline{\mathfrak{q}_2} \in \operatorname{Spec} \overline{B}$. s.t $\overline{\mathfrak{q}_2} \cap A = \overline{\mathfrak{p}_2}$. Then lift back $\overline{\mathfrak{q}_2}$

We have $\mathfrak{q}_2 = \pi^{-1}(\overline{\mathfrak{q}_2})$ $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$ ▨

3. Going down Property.

Prop. VII.3.1. (Local property): Let A be an integral domain. TFAE:

i) A is integrally closed

ii) $A_{\mathfrak{p}}$ is ... $\forall \mathfrak{p}$ prime

iii) $A_{\mathfrak{m}}$ is ... $\forall \mathfrak{m}$ maximal

Lemma: $A \subseteq C \subseteq B$. C is the integral closure of A in B. Then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$

Pf: $S^{-1}C$ integral over $S^{-1}A$, let $\frac{x}{s} \in S^{-1}B$ integral over $S^{-1}A$

Then $(\frac{x}{s})^n + \frac{a_1}{s_1}(\frac{x}{s})^{n-1} + \cdots + \frac{a_n}{s_n} = 0 \Rightarrow (s_1 s_2 \cdots s_n x) \in C \Rightarrow \frac{x}{s} \in S^{-1}C$ ▨

Back to the proof: $K = \operatorname{Frac}(A)$, C be the integral closure of A in K.

$f: A \to C$ $x \mapsto x$ Then A integrally closed $\Leftrightarrow f$ surjective.

$A_{\mathfrak{p}}$ ($A_{\mathfrak{m}}$) integrally closed $\Leftrightarrow f_{\mathfrak{p}}$ ($f_{\mathfrak{m}}$) surjective. ▨

Def VII.3.2 $A \subseteq B$. $\mathfrak{a} \subseteq A$ ideal, $x \in B$ integral over $\mathfrak{a}$ iff $\exists a_i \in I$. s.t. $x^n + a_1 x^{n-1} + \cdots + a_n = 0$.

Lemma VII.3.3. $\mathfrak{a} \subseteq A \subseteq C \subseteq B$. C be the closure, Then the closure of $\mathfrak{a}$ in $B = \sqrt{\mathfrak{a}^e}$, $\mathfrak{a}^e : A \to C$.

Pf: ① $x \in B$ integral over $\mathfrak{a}$. $\Rightarrow$ $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ $a_i \in \mathfrak{a} \subseteq C$.

$\Rightarrow$ $x \in C$ $x^n \in \mathfrak{a}^e \Rightarrow x \in \sqrt{\mathfrak{a}^e}$

② $x \in \sqrt{\mathfrak{a}^e}$, $\Rightarrow x^n \in \mathfrak{a}^e$. that is $x^n = \sum_{i=1}^{m} a_i x_i$ $a_i \in \mathfrak{a}$, $x_i \in C$

where x_i integral over A. $\Longrightarrow$. $A[x_1, \ldots, x_m]$ is a finite generated A-algebra M.

and $x^n M \subseteq \mathfrak{a} M$

Consider $\phi: M \to M$ $m \mapsto x^n \cdot m$. By Caley-Hamilton thm, $\phi(x^n)^r + b_1(x^n)^{r-1} + \cdots + b_r = 0$

in $\operatorname{End}_A(M)$ $\phi(M) \subseteq \mathfrak{a} M \Rightarrow (x^n)^r + b_1(x^n)^{r-1} + \cdots + b_r = 0$. in A $(b_i \in A)$

$\Rightarrow x^n$ integral over $\mathfrak{a}$ $\Rightarrow x$ integral over $\mathfrak{a}$ ▨

Prop VII.3.4 $A \subseteq B$ integral domains A integrally closed, and let $x \in B$ be integral over A.

Then x is algebraic over Frac$(A) = K$. ①

And if $Irr_K(x) = t^n + a_1 t^{n-1} + \cdots + a_n \in K[t]$, then $a_i \in \sqrt{\mathfrak{a}}$ ②

(Particularly, $\mathfrak{a} = (1) \Rightarrow a_i \in A$)

pf ① obvious; ②: L be the splitting field of $Irr_K(x)$ over K. $L = K(x_1, \ldots, x_n)$. (**)

x_i integral over A $(\forall \sigma \in Gal(\bar{K}/K),\ \sigma|_K = id,\ \sigma|_{(x_1,\ldots,x_n)} \subseteq S_n)$

By Vieta thm. a_i are polynomials in x_i. $\overset{(3.3)}{\Rightarrow}$ a_i integral over A

. Since A integrally closed $\overset{(3.3)}{\Rightarrow} a_i \in \sqrt{\mathfrak{a}}$. ▨

Thm (Going-down thm). $A \subseteq B$, B integral over A., A integral closed.

let $\mathfrak{q}_1 \supsetneq \cdots \supsetneq \mathfrak{q}_m \in \operatorname{Spec} B$

$\mathfrak{p}_1 \supsetneq \cdots \supsetneq \mathfrak{p}_m \supsetneq \cdots \supsetneq \mathfrak{p}_n \in \operatorname{Spec} A$ $(n > m)$.

s.t. $\mathfrak{q}_i \cap A = \mathfrak{p}_i$. $(1 \le i \le m)$

Then $\exists\ \mathfrak{q}_{m+1} \supsetneq \cdots \supsetneq \mathfrak{q}_n$ s.t. $\mathfrak{q}_1 \supsetneq \cdots \supsetneq \mathfrak{q}_n$. $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ $(\forall 1 \le i \le n)$

pf: $m = 1$, $n = 2$. find $\mathfrak{q}_2 \subsetneq \mathfrak{q}_1$ s.t. $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$

$\mathfrak{p}_2$ $A \xrightarrow{f} B$ $\mathfrak{p}_2 B$ We prove that: $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A = \mathfrak{p}_2$

β. $\forall x \in B_{\mathfrak{q}_1}\mathfrak{p}_2$. where $x = \frac{y}{s}$. $y \in \mathfrak{p}_2 B$, $s \in B - \mathfrak{q}_1$

$B_{\mathfrak{q}_1}$ $\mathfrak{p}_2 B_{\mathfrak{q}_1}$

Then y is integral over $\mathfrak{p}_2$. (3.3). then its minimal equation over Frac(A) is

$y^r + u_1 y^{r-1} + \cdots + u_r = 0$. $(u_i \in \mathfrak{p}_2)$

$\forall x \in B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A$, Then $s = yx^{-1}$ with x^{-1} in K $\Rightarrow s^r + v_1 s^{r-1} + \cdots + v_r = 0$ $v_i = \frac{u_i}{x^i}$

$\Rightarrow x^i \cdot v_i \in \mathfrak{p}_2$.

Since s integral over A, By (**) we have $v_i \in A$.

If $x \notin \mathfrak{p}_2 \Rightarrow v_i \in \mathfrak{p}_2 \Rightarrow s^r \in B\mathfrak{p}_2 \subseteq B\mathfrak{q}_1 \subseteq \mathfrak{q}_1 \Rightarrow s \in \mathfrak{q}_1$ contradiction

$\Rightarrow x \in \mathfrak{p}_2$. $\Rightarrow$ $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A = \mathfrak{p}_2$.

Now take $\mathfrak{q}_2 = \beta^{-1}(B_{\mathfrak{q}_1}\mathfrak{p}_2)$ we have $\mathfrak{q}_2 \subsetneq \mathfrak{q}_1$, $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$ ▨